

LLM SYSTEMS & LIFECYCLE

LLM Fine-tuning

Model Evaluation

Benchmarking

LoRA / QLoRA

PEFT

Unsloth

vLLM

Transformers

Structured Outputs

Multi-task Training

ML & DEEP LEARNING

Python

PyTorch

TensorFlow

Keras

Scikit-learn

XGBoost

Pandas

NumPy

NLP

Computer Vision

Autoencoders

Feature Engineering

Optimization

ML PLATFORMS & MLOPS

Serving & Orchestration

OpenAI-compatible APIs

Kubernetes / GKE

HPA

Istio

Docker

Seldon Core

Distributed & Cloud

GCP Vertex AI

Dataflow

Apache Beam

Pub/Sub

Azure ML

Lifecycle & Observability

MLflow

Model Registry

Prometheus

Grafana

Cloud Build

ArgoCD

CERTIFICATIONS

Oracle Cloud Infrastructure 2024
Generative AI Certified Professional

Google Cloud
Certified Generative AI Leader

Microsoft Certified
Azure AI Engineer Associate

Microsoft Certified
Azure Data Scientist Associate

EDUCATION

B.E. Electrical & Electronics
Andhra University, Visakhapatnam
2012 - 2016 · 80%

Lead AI and ML engineer and data scientist with **10+ years** of experience, building production AI/ML/LLM systems. Leads end-to-end AI delivery, combining hands-on model development with technical leadership to turn business problems into scalable, high-throughput, low-latency production systems. Experience spans predictive modelling, forecasting, operations research, computer vision, and generative AI across retail, cybersecurity, manufacturing, insurance, and supply-chain optimization. Selected contributions include modelling on **billion-row security datasets**, supply-chain optimization generating **~\$2.5M** in savings, and self-hosted LLMs saving an estimated **~\$300K** annually while improving availability during peak demand.

WORK EXPERIENCE

- **Lead Artificial Intelligence Engineer**
Best Buy · Bengaluru, India
Jul 2025 - Present

- Architected a config-driven Python **LLM training and evaluation platform** spanning data preparation, LoRA/QLoRA, multi-task training, model resolution, accuracy/confusion-matrix/MSE checks, structured-JSON validation, and artifact management.
 - Built reusable LLM fine-tuning, evaluation, and shared vLLM/GKE serving with estimated annual API-cost savings of **~\$300K**. Kept inputs company-managed and reduced external API exposure during peak demand.
 - Built fraud systems across batch, streaming, and real-time models. Beam/Dataflow feature pipelines handled **40,000 events/sec** at peak; supported a **200 ms scoring SLA** with secure, monitored GKE/Seldon Core inference and load testing.
 - Lead architecture and production delivery across search, personalization, fraud, and cybersecurity; partner with data scientists to turn validated models into secure, monitored services.
- **Lead ML Engineer - ML Engineering & Operations**
Wipro · Client: Major US Retail Corporation
Aug 2023 - Jul 2025

- Led ML/MLOps engineers and full model lifecycles on **NVIDIA DGX (8×A100)**, training per-entity autoencoders on billion-row CloudTrail datasets with evaluation, continuous retraining, MLflow, debugging and Grafana alerting. Completed **300+ résumé screens** and **50+ technical interviews** at Wipro.
 - Launched phishing classification at **2,400 emails/sec - 2× prior throughput** and an OCR-based abuse-inbox workflow that doubled case volume, integrating predictions with Microsoft Sentinel and security-response teams.
- **Senior Data Scientist — Data Science / Azure ML**
Accenture
May 2021 - Sep 2023

- Led architecture, experimentation and launch of an Azure ML quality-prediction product: **\$2.5M annual savings**, inspections **130 → 160/day**, **+17%** defect detection and **+20%** customer approval.
 - Owned data preparation, model optimization, API endpoints, automated retraining, CI/CD and monitoring; mentored engineers and aligned business, data science, platform and client stakeholders.

GPU & DISTRIBUTED COMPUTING

CUDA GPU Libraries

GPU-Accelerated Computing

RAPIDS cuDF / CuPy

OPTIMISATION & SIMULATION

Google OR-Tools

Vehicle Routing

Mixed-Integer Programming

Constraint Modelling

Operations Research

Supply-Chain Optimization

MODEL LIFECYCLE

Experiment Tracking

Automated Retraining

Vertex AI Pipelines

Azure ML Pipelines

Vertex AI Feature Store

CLOUD SERVICES

Cloud Run

Cloud Functions

Azure Function Apps

Logic Apps

Google KCC

PRODUCTION ENGINEERING

ML System Architecture

Cloud-Native ML Systems

GitOps

JWT Authentication

Load Testing

Unit / Integration Testing

LANGUAGES & DATA

SQL

BigQuery

MS SQL Server

Redis Caching

GCS

Azure Blob Storage

Artifact Registry

Dask

Git

Linux / Shell Scripting

ENGINEERING LEADERSHIP

Architecture Reviews

Technical Mentoring

Technical Hiring

Team Building

Stakeholder Alignment

Cross-Functional Delivery

EARLIER EXPERIENCE

Data Scientist

GD Research Centre Pvt Ltd

May 2020 - May 2021

Designed and deployed Python forecasting pipelines across 15+ industry verticals, from heterogeneous data preparation through client-facing results.

Deep Learning Engineer

Tata Consultancy Services

Jul 2019 - Mar 2020

Built and deployed an end-to-end semiconductor ML system: distributed training on **~1 million images in ~3 hours; 90% classification accuracy**. Deployed at Amkor, South Korea, and trained the client team.

Senior Software Engineer - ML Infosys

May 2016 - Jul 2019

Delivered design, coding, unit/integration testing, and production support; transitioned to Python ML in 2018 and shipped lead-conversion prediction.

PROJECTS

LLM Fine-tuning, Evaluation & Serving Platform

Best Buy · Jul 2025 - Present

Config-driven Python system for prompt/task configuration, data preparation, LoRA/QLoRA fine-tuning, multi-task training, automated evaluation, structured-output validation, artifact management and vLLM serving. Supports Unsloth and PEFT/Transformers with Accelerate.

GKE

~\$300K estimated annual API-cost savings

Real-time Distributed ML Platform

Best Buy · Aug 2025 - Jan 2026

Beam/Dataflow pipelines transform clickstream and transaction events into Vertex AI Feature Store signals. XGBoost models deploy through Seldon Core on GKE with JWT security, load testing, GitOps releases and production monitoring.

Vertex AI

40k feature events/sec · 200 ms scoring SLA · Production lifecycle ownership

AWS Behavioral Anomaly Models

Wipro · Jan 2024 - Aug 2025

Per-entity autoencoders learn normal AWS account, user, service and machine behavior. Morpheus/RAPIDS training and inference run on DGX over billion-row CloudTrail datasets with continuous retraining, MLflow and Grafana alerts.

NVIDIA Morpheus

RAPIDS / cuDF

Billion-row datasets · Continuous evaluation · 8×A100 DGX

Azure ML Product Quality Platform

Accenture / BamaGruppen · May 2021 - Dec 2021

Led an end-to-end ML product from data preparation and model optimization through API deployment, automated retraining, CI/CD, monitoring, client demonstrations and ongoing production support on Azure.

Function Apps

CI/CD

Monitoring

\$2.5M/year · +17% detection · +20% approval · Launched product

Proofpoint Phishing Classification

Wipro · Oct 2023 - Feb 2024

Built Morpheus training and inference pipelines for preprocessing, feature engineering, model evaluation and live scoring. Integrated security-team feedback and Microsoft Sentinel data to detect phishing that bypassed initial filters.

RAPIDS

DGX

Microsoft Sentinel

Evaluation

2,400 emails/sec · 2× prior throughput · Production security workflow

Semiconductor Computer Vision Platform

TCS / Amkor Technology · Jul 2019 - Mar 2020

Built and deployed an end-to-end semiconductor ML system: distributed training on **~1 million images in ~3 hours**, with **90% classification accuracy**. Deployed at Amkor, South Korea, and trained the client team.

TensorFlow / Keras

DenseNet

Distributed GPU

~1M images · ~3-hour training cycles · Production launch

PLATFORM ENGINEERING EVIDENCE

vLLM Model Chooser

Live tool · ylnhari.github.io/vllm-model-chooser

Selects a vLLM-compatible model for a target GPU by VRAM, quantization and KV-cache fit across 115 model configurations.

GitOps Delivery Platform

Best Buy · Cloud Build · ArgoCD · Google KCC

Reusable delivery standards for models, pipelines and GCP resources, with declarative infrastructure and zero-downtime releases to GKE.

Rover

Go · github.com/ylnhari/rover

Remote command executor and project launcher written in Go: single static binary, zero dependencies and real-time SSE streaming.